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Music Recommendation with Large Language Models: Challenges, Opportunities, and Evaluation

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Music Recommender Systems (MRS) have long relied on an information-retrieval framing, where progress is measured mainly through accuracy on retrieval-oriented subtasks. While effective, this reductionist paradigm struggles to address the deeper question of what makes a good recommendation, and attempts to broaden evaluation, through user studies or fairness analyses, have had limited impact. The emergence of Large Language Models (LLMs) disrupts this framework: LLMs are generative rather than ranking-based, making standard accuracy metrics questionable. They also introduce challenges such as hallucinations, knowledge cutoffs, non-determinism, and opaque training data, rendering traditional train/test protocols difficult to interpret. At the same time, LLMs create new opportunities, enabling natural-language interaction and even allowing models to act as evaluators.

This work argues that the shift toward LLM-driven MRS requires rethinking evaluation. We first review how LLMs reshape user modeling, item modeling, and natural-language recommendation in music. We then examine evaluation practices from NLP, highlighting methodologies and open challenges relevant to MRS. Finally, we synthesize insights—focusing on how LLM prompting applies to MRS—to outline a structured set of success and risk dimensions. Our goal is to provide the MRS community with an updated, pedagogical, and cross-disciplinary perspective on evaluation.

Additional Key Words and Phrases: Music Recommender Systems, Item Modeling, User Modeling, Natural Language Requests, LLMs, Evaluation, Metrics, Risks, Challenges

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1 Introduction

At its origins, the field of Music Recommender Systems (MRS) was built upon the same foundations as the broader Recommender Systems (RS) domain, namely, by framing its objectives as a set of information retrieval tasks. Practitioners adopting this view benefited from the rapid advancements in the RS community and in machine learning in general. For decades, progress in MRS essentially meant improving performance on a series of retrieval-oriented subtasks, often

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evaluated offline, such as matrix completion, clustering, or sequence continuation. Soon enough, this framing revealed its limitations due to its markedly reductionist nature. It simplified the original question, *how do we design systems that make good recommendations?*, by sidestepping the harder one: *what actually constitutes a good recommendation?* Instead, the focus shifted to a more tractable problem: *how can we build systems that accurately predict what users will eventually consume?* While the two are arguably not the same, the latter offered a more straightforward way to evaluate the proposed systems in terms of algorithmic performance. And, as often happens in scientific practice, the measure gradually became the objective [19].

Attempts to bridge the gap between the evaluation paradigm and the original question have largely focused on introducing user studies that go beyond accuracy-based evaluation [79, 139], or on addressing important adjacent issues such as biases, fairness, and algorithmic accountability [36]. Nonetheless, the underlying paradigm has remained largely unchanged for decades, and RS papers continue to be evaluated primarily through accuracy metrics on standard benchmark datasets. Although critical perspectives have been generally welcomed within the community [28], they have rarely succeeded in fundamentally reshaping this evaluation framework. Put simply, in the absence of a clear and practical alternative, the simplistic accuracy-driven paradigm has persisted.

The emergence of Large Language Models (LLMs) and their rapid adoption in RS disrupts this long-standing evaluation practices, as the fundamental assumptions of this established paradigm no longer hold [31, 32]. LLMs are generative models—they do not compute similarity scores nor rank items in the conventional sense—and therefore cannot be meaningfully evaluated using standard accuracy metrics. More critically, such metrics were not designed to account for hallucinations, knowledge cutoffs, the influence of training data sources, or personalization. Moreover, the very notion of train/test separation becomes largely irrelevant for black-box models such as ChatGPT—especially given that these models have likely been exposed to the public datasets used for evaluation, making reproducibility and fair comparison of results difficult, if not impossible.

At the same time, LLMs have inspired a new wave of methodologies and can even be applied directly to the core MRS objective, that is, users can simply ask an LLM for music recommendations, much as they would ask a human acquaintance, or they could ask LLMs to assess recommendations in a role as judges.

Despite the breadth of challenges that LLMs introduce to traditional MRS evaluation, many studies have continued to rely on standard accuracy-based metrics, often without fully considering what these numbers signify for models that generate text rather than predict or rank discrete items. Regardless of where one stands in the debate over the disruptive nature of LLMs, their growing use in MRS represents a valuable opportunity to revisit how we evaluate recommender systems in general. It encourages the exploration of new evaluation perspectives that better align with the human experience of receiving music recommendations, while also prompting reflection on what constitutes a poor recommendation in generative settings, such as cases of hallucination or factual inconsistency. Effectively, it also requires taking inspiration from the field of Natural Language Processing (NLP) to evaluate the text produced by LLMs.

This paper is structured as follows: First we provide a comprehensive overview of how LLMs have transformed various aspects of user modeling, item modeling, and recommendation, particularly in the music domain. Then we discuss evaluation at the general level, by considering each scenario as a text generation task and exposing what is done in adjacent research domains. Finally, we draw inspiration from existing work in music recommendation to outline current strategies for measuring both success and risks, while introducing a detailed set of evaluation dimensions and challenges that we argue should be considered in any study involving LLMs as recommenders.

An overview of the paper content is presented in Figure 1. Overall, our contributions are:

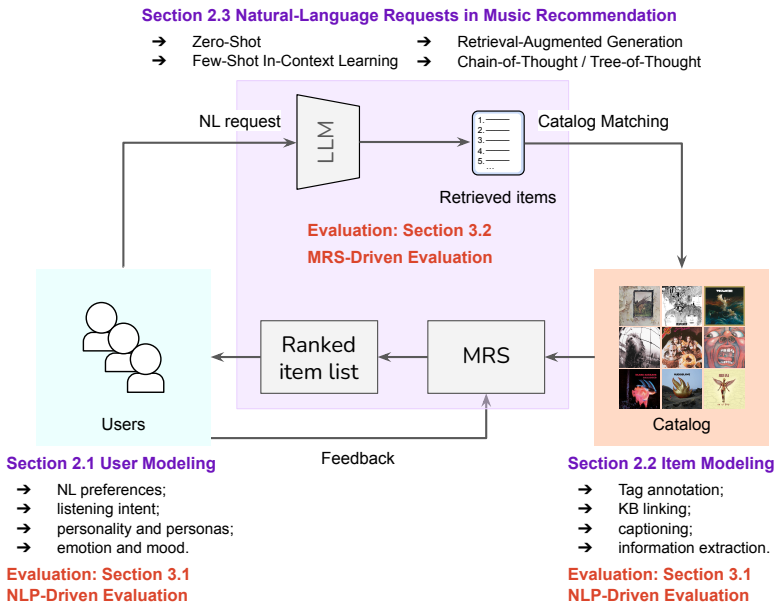

Fig. 1. Paper's overview as a generic diagram presenting music recommendation with LLMs

- A detailed overview of background, challenges, and LLM-induced changes in MRS in Section 2 with an emphasis on the specificity of music consumption;
- An evaluation framework inspired from NLP works that goes beyond traditional IR metrics and covers a wider range of scenarios: human vs. automatic, reliance on ground-truth or not, and so on (Section 3.1);
- A comprehensive catalog of success and risk evaluation dimensions specific to the assessment of LLM-based MRS (Section 3.2).

2 LLMs in Music Recommendation: Domain-specific Challenges and Opportunities

In the following, we focus on three key aspects of music recommender systems where LLMs have been applied: item modeling, user modeling, and natural language-based music recommendation. For each of these aspects, we structure the discussion around three main points: 1) Background or fundamentals of the task; 2) Music-specific challenges and domain considerations, highlighting what makes the problem unique in the context of music. 3) Impact of LLMs, analyzing how generative language models can or have already changed methods and capabilities in the respective task.

Past research on music item or user modeling for MRS has primarily focused on two directions: (1) enriching items/users with explicit, often high-level information (e.g., tag annotation or extraction, knowledge base linking) and (2) representing item/users as meaningful features for use in content-based or hybrid recommender systems (e.g., representation learning.) In what follows, we concentrate on the first aspect, which directly leverages the generative capabilities of LLMs to enrich the item or user information. The second aspect, involving text embedding and feature representation, is left for future work, as it encompasses broader approaches that extend beyond LLMs to include other text encoders and representation models.

2.1 User Modeling

A first use case of LLMs in the context of MRS is to create and extend user profiles in an accurate and comprehensive way, tailored to the music domain and recommendation task at hand. In the following, we first describe LLMs' potential for general music preference modeling, pertaining to traditional information such as consumption or interaction histories. Subsequently, we will discuss their merits for advanced user modeling, involving personas, affective cues, listening intents, and cognitive effects.

Background. Most automatic music recommender systems rely on modeling user preferences. These preferences are often represented as vectors or embeddings, learned from interaction data using neural architectures in collaborative filtering settings [63, 66, 77]. A key limitation of such embeddings, derived from black-box models, is their opacity: they are not easily interpretable. This hinders both the generation of explanations for users and the investigation of the underlying decision processes. For user representations to be effective, a comprehensive collection of interaction data is required, linking each user to a diverse set of items. This requirement introduces several drawbacks. For instance, cold-start users, those who have just joined the platform, lack a meaningful representation and therefore receive poor recommendations. Similarly, if a user wishes to steer their recommendations toward new interests, the system typically demands a substantial number of interactions with items from these new domains before the model can adapt.

User profiles can also span to the level of *personas*, i.e., groups of users that share common characteristics or tastes. These can be defined, among others, at the level of demographics (age, gender, place of residence), involve cultural aspects (religion, beliefs, norms), or socio-economic factors (wealth, income, social class). Together with models that connect one or more of these factors to inclinations of music preference, e.g. [90, 182], such personas have been used to overcome cold-start scenarios. For instance, the MRS can only use the user's belonging to a demographic group, to make recommendations, in a traditional demographic filtering approach. Alternatively, the definition of personas can also more closely reflect certain music-related patterns or dispositions. For instance, personas can be modeled as having different preferences for musical diversity, popularity, or mainstreaminess. [9, 83].

Several works have provided evidence for a strong connection between music preferences and *psychological aspects*, including *personality* [103, 137], *listening functions* and *intent* [94, 140] as well as *affect* [51, 113, 116]. The purposes or reasons why people listen to music have been studied intensively in music psychology literature [94, 140], resulting in the emergence of three major trajectories, i.e., mood regulation, self-awareness, and social relatedness.

Personality, often represented via the Big Five Model (aka Five Factor Model), describes traits along the dimensions of openness, conscientiousness, extraversion, agreeability, and neuroticism [101]. While personality can also be part of a persona description, it goes beyond common persona definitions, especially in the context of music because of its nuanced relationship with musical perception and preferences. For instance, correlations have been identified between personality traits and acoustical properties such as loudness or tempo [103], genre preferences [129], and perceived emotion [137]. Based on such insights, a few attempts to integrate personality information into the music recommendation process have been made [92, 95, 149]. Furthermore, modeling affective states of listeners is strongly related to modeling *listening intents*, because mood regulation is one of the dominant reasons why humans listen to music [140]. Eliciting affective cues such as *emotions* (short-lasting reactions in response to a stimulus), *mood* (longer-lasting episodes independent of a concrete stimulus), or *sentiment* (general positive or negative attitude towards a concrete item) is therefore important to tailor recommendations to the user's current affective state.